Rolling Element Bearing Diagnostics: Envelope Demodulation and Spectral Analysis

Section 1: Physical Mechanism of Localised Bearing Defects

When a rolling element strikes a localised spall on the inner or outer raceway, it produces a short-duration broadband impulse that excites structural resonances of the bearing assembly and transducer mounting (commonly 2000 to 6000 Hz). Because rolling elements experience random slip of 1 to 2 percent under load, high-order harmonics in the direct baseband FFT smear out, whereas demodulating a band-pass filtered resonance band via the Hilbert transform (envelope spectrum) extracts the repetition rate (BPFO, BPFI, BSF, FTF) with high signal-to-noise ratio.

Section 2: Why a Single FFT Peak Does Not Confirm a Bearing Fault

An isolated peak in the raw FFT spectrum near a theoretical bearing characteristic frequency (FTF, BPFO, BPFI, or BSF) is never definitive proof of a bearing defect. Coincidental peaks frequently arise from electrical line harmonics (50/60 Hz and multiples), gear-mesh or belt pass tones, blade/vane pass frequencies, or shaft harmonic distortion from misalignment and mechanical looseness.

Diagnostic confirmation requires corroborating evidence across multiple indicators: (1) harmonic series (1x, 2x, 3x) in the high-frequency envelope spectrum, (2) modulation sidebands (1x shaft RPM sidebands around BPFI; FTF sidebands around BSF), and (3) elevated impulsiveness in time-domain kurtosis (> 4.0) and crest factor (> 5.0).

Section 3: Time-Frequency Spectrogram and Additional Recommended Measurements
Short-Time Fourier Transform (STFT) spectrograms reveal whether high-frequency energy occurs as periodic transient vertical stripes (characteristic of impacts) or continuous stationary horizontal bands (characteristic of tonal noise or continuous rubbing).
When vibration analysis indicates a potential raceway or rolling-element defect, recommended follow-up measurements include: tri-axial accelerometer measurements (horizontal, vertical, axial), high-frequency envelope trending, phase measurement across couplings to rule out misalignment, infrared thermography of the bearing housing, and oil/grease ferrography for metallic spall particles.